

Exercise sheet 3

Topics in Topology, MTH602

1. Given a covering $p : X \rightarrow Y$, if $B \subset Y$ and $A = p^{-1}(B)$, then prove that the restriction $p : A \rightarrow B$ is also a covering.
2. Find the deck transformation groups of the coverings: $p : \mathbb{R} \rightarrow S^1$, and $p_2 : S^2 \rightarrow \mathbb{RP}^2$.
3. Find *all* the covers of $\mathbb{RP}^2$ and S^1 .
4. Find a simply connected cover of the wedge of two circles. (*Hint:* Try lifting the circles)
5. Find a cover $p : X \rightarrow S^1 \vee S^1$ of the wedge of two circles so that the image of p_* is $\mathbb{Z}$.
6. If X is any topological space with a finite fundamental group, then show that any $f : X \rightarrow S^1$ must be homotopic to the constant map.